Computer Architecture_2024_PYQ

Group-A (Very Short Answer Type Question)

Answer any ten of the following: [1 x 10 = 10]

1. What is meant by data dependence?
2. Which page replacement algorithm suffers from Belady's anomaly?
3. Which architecture is/are suitable for realizing SIMD?
4. What is meant by Branch Prediction?
5. The throughput of a super scalar processor is _____.
6. Which unit is responsible for translation of logical address to physical address?
7. The _____ plays a very vital role in case of super scalar processors.
8. The set of loosely connected computers are called as _____.
9. Write the equation for Amdahl's Law.
10. Write the statement for memory inclusion property.
11. What is meant by instruction level parallelism?
12. In tightly coupled systems, the microprocessors share _____.

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Group-B (Short Answer Type Question)

Answer any three of the following: [5 x 3 = 15]

2. Given a 4-segment pipeline whereby each segment has a delay time as follows:

Segment1: 40 ns, Segment2: 25 ns, Segment3: 45 ns, Segment4: 45 ns.

The delay time for the interface register is 5ns. Calculate:

i) cycle time of the non-pipeline and pipeline,

ii) execution time for 100 tasks,

iii) real speedup,

iv) maximum speed up. [5]

3. In a simple machine with load-store architecture having clock rate of 1.8GHz and the following specifications:

Operations	**Frequency**	**Number of clock cycles**
ALU	40%	1
Load	20%	2
Store	10%	2
Branch	30%	2

Calculate CPI and MIPS rating for the machine. [5]

4. What is a superscalar processor? State the advantages of vector computer. [5]

5. State the differences between static network and dynamic network. Explain the hypercube interconnection with $n = 3$. [5]

6. Compare superscalar, super-pipelined and superscalar-super-pipelined architecture. [5]

Group-C (Long Answer Type Question)

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Answer any three of the following: [15 x 3 = 45]

7.

a. Compare tightly coupled system and loosely coupled system. [6+9]

b. Explain with suitable diagram the multiprocessor architectures (UMA, NUMA, COMA).

8.

a. What is page fault? [2]

b. Given page reference string: [1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6].

Compare the page fault rates for LRU, FIFO and Optimal page replacement algorithm. [9]

c. Discuss the implementation of virtual memory through segmentation with suitable diagram. [4]

9.

a. Discuss SIMD array processor architecture with suitable diagram. [5]

b. What are vector stride and vectorization? [2]

c. What is the difference between scalar processor and vector processor? [2]

d. Explain vector gather and scatter instructions with suitable diagrams. [6]

10.

a. What is cache coherency? [2]

b. Explain the MESI protocol briefly. [3]

c. Explain the snoopy bus protocol for cache coherency. [6]

d. Explain how synchronization is ensured in multiprocessor environment? [4]

11.

a. Explain VLIW architectures with suitable diagram. [6]

b. State the advantages and disadvantages of VLIW architecture. [5]

c. What are the hurdles in superscalar architecture? [4]

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